

ARAVIND KUMAR V

B.Tech in Artificial Intelligence and Data Science | Sri Manakula Vinayagar Engineering College,
Puducherry

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PROFILE

Innovative AI and IoT Developer with strong expertise in Machine Learning, embedded systems, and real-time signal processing. Final year B.Tech student specializing in Artificial Intelligence and Data Science with hands-on experience in developing intelligent IoT solutions for healthcare, agriculture, and assistive technologies. Proficient in Python, TensorFlow, and Arduino-based systems. Passionate about leveraging AI and machine learning to create scalable and impactful solutions. And also with IOT.

EDUCATION

B.Tech in Artificial Intelligence and Data Science (Final Year) | CGPA: 6.88

Sri Manakula Vinayagar Engineering College, Puducherry

Higher Secondary Education (Class 12, State Board) | 2022

Amalorpavam Hr. Sec. School | Percentage: 62%

Secondary Education (Class 10, State Board) | 2020

Amalorpavam Hr. Sec. School | Percentage: 58%

TECHNICAL SKILLS

Programming Languages: Python, C, C++, HTML, CSS

Machine Learning & AI: TensorFlow, Keras, PyTorch, Scikit-learn, CNN, Image Processing, Anomaly Detection.

IoT & Embedded Systems: Arduino IDE, ESP32, Raspberry Pi (Basics), Sensor Integration, Real-time Data Acquisition, Signal Processing, Microcontroller Programming

Data Science Tools: Pandas, NumPy, Matplotlib, Seaborn, Data Preprocessing, Feature Engineering.

Web Development & Deployment: Flask, Web Design, Database Management

CERTIFICATIONS & COURSES

- Introduction to Machine Learning – Comprehensive foundational course
- Python for Data Science – Advanced data science techniques and algorithms
- OOPs Concepts in C++ – Great Learning
- C, C++, Python, HTML, CSS, JavaScript, Excel Certification – Great Learning
- AWS Academy Cloud Foundations – AWS Academy Graduate

PROFESSIONAL EXPERIENCE

IoT + ML Developer Intern | Edusphere Solutions, Puducherry

March 2026 – June 2026

- Developed and deployed real-time IoT and Machine Learning projects using Arduino IDE, ESP32, and Python
- Built intelligent systems integrating sensors, microcontrollers, and ML models for data-driven decision making
- Created web applications using Flask for IoT data visualization, monitoring, and real-time analytics
- Implemented end-to-end AI/ML solutions from data collection to model deployment on IoT devices

Python Developer Intern | Edusphere Solutions, Puducherry

September 2025 – December 2025

- Developed Python-based applications and scripts for data processing and analysis
- Implemented data preprocessing pipelines and feature engineering techniques
- Collaborated with team members on data-driven projects and contributed to codebase development

KEY PROJECTS

Real-Time ECG, EMG, EOG Signal Acquisition and Anomaly Detection Using BioAmp EXG and Machine Learning

- Engineered real-time biomedical signal acquisition system using BioAmp EXG sensor integrated with ESP32/Arduino for continuous health monitoring
- Implemented advanced signal conditioning techniques (filtering, amplification, digitization) to ensure data accuracy
- Developed TensorFlow-based ML models for ECG, EMG, and EOG anomaly detection, achieving real-time classification of heart conditions and muscle disorders
- Created healthcare monitoring system with real-time alerts for early diagnosis and patient care enhancement
- *Technologies: Python, TensorFlow, Keras, Arduino IDE, ESP32, Signal Processing, Pandas, NumPy*

EOG Wheelchair with Fall Detection System

- Designed assistive wheelchair system controlled by Electrooculography (EOG) signals for hands-free navigation of physically disabled individuals
- Implemented real-time eye movement detection (left, right, blink) with signal processing algorithms mapped to wheelchair directions
- Integrated accelerometer and gyroscope sensors for advanced fall detection with automated emergency notifications to caregivers
- Enhanced user safety and independence for individuals with severe mobility impairments
- *Technologies: Arduino, ESP32, EOG Sensors, Accelerometer, Gyroscope, Python, Signal Processing*

EMG-Based Wheelchair Control System

- Developed intelligent wheelchair control system using Electromyography (EMG) signals from muscle contractions
- Applied advanced signal conditioning techniques (amplification, noise filtering) for reliable EMG signal capture
- Implemented ML classification algorithms to map muscle signals to wheelchair commands (forward, backward, left, right)

- Created cost-effective, user-friendly solution for individuals unable to use traditional joystick-based controls
- *Technologies: Arduino IDE, ESP32, EMG Sensors, Signal Processing, Python, Machine Learning*

Smart Agriculture with Wildlife, Leaf Disease, and Animal Detection using AI

- Developed comprehensive IoT and AI-based precision agriculture platform with real-time crop monitoring and threat detection
- Implemented deep learning models (CNN) for early leaf disease identification, enabling timely intervention and crop protection
- Deployed computer vision for wildlife and animal intrusion detection with automated alert and deterrent systems
- Integrated environmental sensors for soil and climate monitoring to optimize irrigation and crop management decisions
- Reduced crop damage and increased agricultural productivity through intelligent, automated farming solutions
- *Technologies: Python, TensorFlow, OpenCV, Deep Learning, IoT Sensors, Image Processing, Computer Vision*

Mini Projects: Airbag System (collision detection), Anemia Detection System (sensor-based health monitoring), Automatic Railway Gate Control (IoT automation), Alcohol Detection System for Cars (safety application)

ACHIEVEMENTS & PARTICIPATION

- **Datathon Participant – SRM University (February 25, 2025):** Developed advanced ML models with focus on data cleaning, feature engineering, and preprocessing for competitive data science challenges

SOFT SKILLS

Leadership, Problem-Solving, Communication, Teamwork, Self-Learning, Research-Oriented.

LANGUAGES

English (Fluent), Tamil (Fluent)